Achromatopsia: Clinical Trials, Molecular Mechanisms and Current Research

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1. Disease Background

Achromatopsia (ACHM) is a rare autosomal recessive inherited retinal disease affecting approximately 1 in 30,000–50,000 individuals (Kohl *et al.*, 2005; Remmer *et al.*, 2015). It is characterised by the complete or near-complete absence of cone photoreceptor function from birth, causing total colour blindness, severely reduced visual acuity (typically worse than 20/200), extreme photophobia, pendular nystagmus and a central scotoma (Kohl *et al.*, 2005). Rod-mediated (scotopic) vision is largely preserved. Although historically considered a stationary disease, recent longitudinal studies have shown progressive decline in photoreceptor structural integrity — macular thinning, disruption of the ellipsoid zone (EZ) and RPE atrophy — in many patients over time (Aboshiha *et al.*, 2014; Brunetti-Pierri *et al.*, 2021; Rohowetz *et al.*, 2025). This implies a finite therapeutic window: gene therapy is likely more effective before irreversible photoreceptor loss accumulates.

Genetic Architecture

Achromatopsia is caused by loss-of-function mutations in six genes encoding proteins in the cone phototransduction cascade (Baxter and Borchert, 2024; Kohl *et al.*, 2005):

- **CNGB3** — 50–70% of cases; most prevalent in European populations (Kohl *et al.*, 2005). Encodes the beta-subunit of the cone cyclic nucleotide-gated (CNG) channel.
- **CNGA3** — ~25% of cases; more prevalent in Asian and Middle Eastern populations (Sun *et al.*, 2020). Encodes the alpha-subunit of the cone CNG channel.
- **ATF6** — ~2%; encodes activating transcription factor 6, a regulator of the unfolded protein response (Ansar *et al.*, 2015).
- **GNAT2** — <2%; encodes cone-specific transducin alpha subunit.
- **PDE6C** — <2%; encodes catalytic alpha subunit of cone phosphodiesterase 6.
- **PDE6H** — 0.3%; encodes inhibitory gamma subunit of cone PDE6 (Baxter and Borchert, 2024).

CNGB3 and CNGA3 together account for ~75–80% of all cases (Kohl *et al.*, 2005), making them the dominant targets for current gene therapy development.

2. The Cone Phototransduction Cascade — and Where It Fails in Achromatopsia

In darkness, cone photoreceptors maintain a tonically depolarised state (approximately –40 mV) driven by a high intracellular concentration of cyclic guanosine monophosphate (cGMP), which holds the CNG channels open. The cone CNG channel is a heterotetrameric integral membrane protein composed of three CNGA3 (alpha) subunits and one CNGB3 (beta) subunit, located on the plasma membrane of the cone outer segment (Kohl *et al.*, 2005; Baxter and Borchert, 2024). Open CNG channels allow a continuous inward cation current (the 'dark current', primarily Na+ and Ca2+) that sustains glutamate release from the cone synaptic terminal onto downstream bipolar cells.

Light Response: Signal Initiation to Hyperpolarisation

Photon absorption by cone opsin causes photoisomerisation of 11-*cis* retinal to all-*trans* retinal, activating opsin. Activated opsin catalyses GDP→GTP exchange on the cone-specific G protein transducin alpha subunit (GNAT2), causing it to dissociate and activate cone phosphodiesterase 6 (PDE6, composed of catalytic PDE6C and inhibitory PDE6H subunits). PDE6 hydrolyses cGMP at a high rate, rapidly reducing intracellular cGMP. As cGMP falls, CNG channels close — terminating the dark current. Reduced Ca2+ entry (combined with ongoing Ca2+ extrusion by NCKX exchangers) causes intracellular Ca2+ to fall. The net effect is membrane hyperpolarisation to approximately –70 mV, reducing synaptic glutamate release and producing a signal processed by the retinal circuitry and ultimately relayed to the visual cortex (Baxter and Borchert, 2024).

Recovery and Adaptation

Recovery involves opsin phosphorylation (GRK1) and arrestin binding, GTP hydrolysis by the Gt-alpha subunit (accelerated by RGS9), cGMP resynthesis by retinal guanylyl cyclase (GC), and Ca2+-feedback through guanylyl cyclase activating proteins (GCAPs) — restoring the dark state. This cycle underlies the capacity for light and dark adaptation.

How CNG Channel Mutations Cause Achromatopsia

Pathogenic CNGB3 or CNGA3 mutations prevent the CNG channel from assembling correctly, trafficking to the outer segment membrane, or functioning as an ion channel. Without the dark current, the cone is chronically hyperpolarised and cannot signal changes in light intensity — it is functionally silent. Critically, CNGB3 mutations also cause secondary mislocalization of CNGA3 protein, since the two subunits depend on each other for correct outer segment trafficking (Kohl *et al.*, 2005; Baxter and Borchert, 2024). Despite functional silence, cone photoreceptors may be structurally preserved (visible on OCT as intact ellipsoid zone) in younger patients, providing the anatomical substrate for gene therapy (Genead *et al.*, 2011).

The ATF6/ER Stress Pathway in Achromatopsia

ATF6-associated achromatopsia (~2% of cases) operates through a distinct mechanism. ATF6 is a type II ER-resident transmembrane protein that is activated by ER stress: when misfolded proteins accumulate, the chaperone GRP78/BiP releases ATF6, which translocates to the Golgi, is cleaved by S1P/S2P proteases, and its N-terminal fragment acts as a nuclear transcription factor upregulating ER chaperones and ERAD components — one arm of the unfolded protein response (UPR). Without functional ATF6, cones cannot manage their high proteostatic demand, leading to ER stress, dysfunction and degeneration (Ansar *et al.*, 2015; Baxter and Borchert, 2024).

3. AAV Gene Therapy — Complete Molecular Mechanism

Adeno-associated virus (AAV) is the delivery vehicle of choice for retinal gene therapy. It is a ~25 nm non-enveloped parvovirus with a single-stranded DNA (ssDNA) genome. Recombinant AAV (rAAV) vectors retain only the ~145 bp inverted terminal repeats (ITRs) from the wild-type genome — all viral coding sequences are replaced by the therapeutic cassette — making rAAV replication-defective and non-pathogenic (Baxter and Borchert, 2024). The following describes every step from vector design through to restored cone function.

Step 1 — Vector Construction

For achromatopsia, the rAAV vector contains: (1) AAV serotype 8 (or modified AAV2) capsid; (2) AAV2 ITRs as *cis*-acting packaging and second-strand synthesis signals; (3) a cone-specific promoter (hCAR, ARR3, or PR1.7 depending on the trial); (4) the full-length human CNGB3 or CNGA3 cDNA; (5) in some constructs, a mutated WPRE for enhanced mRNA stability; and (6) a polyadenylation signal. The cone-specific promoter is essential: it contains transcription factor binding motifs recognised exclusively in cone photoreceptors, restricting expression to the therapeutic target cells and avoiding ectopic channel expression in rods, RPE or Müller glia (Baxter and Borchert, 2024; Carvalho *et al.*, 2011).

- The **hCAR (human cone arrestin) promoter** is a ~290 bp fragment used in MeiraGTx's CNGB3 and CNGA3 vectors (Michaelides *et al.*, 2023).
- The **hArr3 (arrestin-3) promoter** is used in the STZ eyetrial CNGA3 vector (Reichel *et al.*, 2022).
- The **PR1.7 promoter** (~1.7 kb red cone opsin promoter fragment) is used in both AGTC-401 and AGTC-402 (AGTC press release, 2022).

Why AAV8 Specifically

AAV8 was selected for most achromatopsia trials because: it transduces cone photoreceptors and RPE more efficiently than AAV1, 2, 4, or 6 after subretinal delivery; it uncoats its genome faster than AAV2 after nuclear entry, producing more rapid transgene expression; and neutralising antibodies to AAV8 are less prevalent in humans than to AAV2, reducing the risk of immune-mediated vector neutralisation (Baxter and Borchert, 2024). AGTC's approach used modified AAV2 — specifically rAAV2tYF, where surface-exposed tyrosine residues are mutated to phenylalanine to block phosphorylation-mediated proteasomal degradation of the capsid, substantially improving nuclear accumulation and transduction efficiency (AGTC press release, 2022).

Step 2 — Subretinal Injection

All achromatopsia gene therapy trials deliver the vector subretinally — not intravitreally. The procedure involves a pars plana vitrectomy, creation of a small retinotomy, and injection of 100–500 µL of vector solution through a fine-gauge cannula into the subretinal space, creating a localised bleb of retinal detachment over the macula where cones are densest. The bleb reabsorbs within hours to days as RPE pump activity restores retinal attachment (Baxter and Borchert, 2024; Reichel *et al.*, 2022). Subretinal delivery is preferred over intravitreal injection because the inner limiting membrane (ILM) constitutes a significant barrier preventing AAV penetration to the photoreceptor layer from the vitreous, and subretinal placement positions the vector in direct contact with cone outer segments and RPE (Baxter and Borchert, 2024).

Step 3 — Receptor Binding

In the subretinal space, AAV8 capsids contact the outer segments of cone photoreceptors. AAV8 uses laminin receptor (LamR) as its primary receptor, with additional binding via heparan sulphate proteoglycans (HSPGs) and galactose — a different receptor profile from AAV2, contributing to AAV8's broader and more efficient photoreceptor transduction (Baxter and Borchert, 2024).

Step 4 — Endocytosis and Endosomal Escape

After receptor binding, the capsid is internalised primarily via clathrin-mediated endocytosis. As the endosome acidifies, a conformational change in the capsid exposes the VP1 N-terminal phospholipase A2 (PLA2) domain. PLA2 activity disrupts the endosomal membrane, enabling capsid escape before lysosomal degradation — a critical rate-limiting step in transduction efficiency (Baxter and Borchert, 2024).

Step 5 — Nuclear Entry and Capsid Uncoating

The intact capsid (~25 nm) traffics through the cytoplasm and enters the nucleus via importin-beta-mediated active transport through nuclear pore complexes. Within the nucleus, the capsid undergoes uncoating — disassembly driven by the nuclear environment and redox changes — releasing the ssDNA genome. AAV8 uncoats substantially faster than AAV2, contributing to its superior expression kinetics (Baxter and Borchert, 2024).

Step 6 — Second-Strand Synthesis and Episomal Persistence

Host cell DNA polymerase delta converts the ssDNA to dsDNA using the ITR hairpin as a self-primer. The resulting dsDNA forms circular episomes that persist extrachromosomally in post-mitotic photoreceptors — which do not divide, so the episome is not diluted — providing long-lasting transgene expression without genomic integration (Baxter and Borchert, 2024; Carvalho *et al.*, 2011). The durability of expression was confirmed in the STZ eyetrial at 3-year follow-up (Reichel *et al.*, 2022).

Step 7 — Transcription and Translation

RNA polymerase II transcribes the therapeutic cDNA under control of the cone-specific promoter. Only cone photoreceptors — which express the appropriate transcription factors for hCAR, ARR3, or PR1.7 activity — produce the mRNA. The WPRE element (where present) enhances mRNA stability and nuclear export. Ribosomes on the rough ER and in the cytoplasm translate the mRNA, and CNGB3/CNGA3 — integral membrane proteins — are co-translationally inserted into the ER membrane via the SRP/translocon pathway, where they are glycosylated and undergo chaperone-assisted folding (calnexin, calreticulin) (Baxter and Borchert, 2024).

Steps 8–10 — Assembly, Trafficking, and Restoration of Dark Current

Following ER quality control, CNGA3 and CNGB3 subunits assemble into their physiological heterotetrameric configuration (three CNGA3 + one CNGB3) in the ER or early Golgi (Kohl *et al.*, 2005). The assembled channel complex is trafficked via vesicular transport along microtubules from the cone inner segment (site of synthesis) to the outer segment membrane — a polarised trafficking step dependent on C-terminal targeting sequences (Baxter and Borchert, 2024). Once inserted in the outer segment plasma membrane, the functional CNG channels respond to cGMP and restore the dark current. With dark current restored, the cone can once again signal changes in light intensity through the full phototransduction cascade — confirmed experimentally by restoration of cone-mediated ERG responses in animal models (Carvalho *et al.*, 2011), and by functional and neuroimaging measures in human trials (Farahbakhsh *et al.*, 2022; Reichel *et al.*, 2022).

> **Timeline:** In animal models, measurable cone ERG responses appear within 4–8 weeks of subretinal injection. In human trials, functional improvements are typically first detectable at 3-month assessments and continue to evolve through 12–24 months, reflecting the cumulative kinetics of second-strand synthesis, transcription, translation, protein trafficking, and ultimately cortical adaptation to restored cone input (Farahbakhsh *et al.*, 2022; Reichel *et al.*, 2022).

4. The Clinical Trials — In Depth

Trial 1: AAV-CNGB3 — MeiraGTx (NCT03001310)

Status: Completed | **Phase:** 1/2 | **Sponsor:** MeiraGTx UK II Ltd (acquired by Janssen/J&J 2023) | **Started:** January 2017

Background and Rationale

The first-in-human (FIH) gene therapy trial for CNGB3-achromatopsia in the UK, led by Prof. Michel Michaelides at UCL/Moorfields Eye Hospital. It built on robust preclinical proof-of-concept: subretinal AAV-CNGB3 delivery had already restored cone ERG responses in CNGB3-mutant dogs and mice (Carvalho *et al.*, 2011; Komaromy *et al.*, 2010).

Vector Design

The vector, **AAV2/8-hCARp.hCNGB3**, uses an AAV8 capsid with AAV2 ITRs, the ~290 bp human cone arrestin (hCAR) promoter driving cone-specific expression, full-length human CNGB3 cDNA, and an SV40 polyadenylation signal (Michaelides *et al.*, 2023).

Trial Design

Prospective, open-label, non-randomised, dose-escalation Phase 1/2 trial enrolling 23 adults and children (≥3 years) with confirmed biallelic CNGB3 mutations. Adults received one of three escalating subretinal doses (up to 0.5 mL) in the worse-seeing eye; a paediatric expansion followed at the maximum tolerated adult dose. Primary endpoint: safety. Secondary endpoints: best-corrected visual acuity, contrast sensitivity, colour vision, photo-aversion, retinal sensitivity and vision-related quality of life (Michaelides *et al.*, 2023).

Results

The trial demonstrated an **acceptable safety profile**. Three of 23 participants (13%) experienced primary safety outcome events — severe intraocular inflammation — requiring corticosteroid treatment; all resolved without permanent visual loss (Michaelides *et al.*, 2023). Efficacy signals:

- **Photo-aversion:** improved in 11 of 20 evaluable patients (Michaelides *et al.*, 2023).
- **Colour vision:** improved in 6 of 23 participants (Michaelides *et al.*, 2023).
- **Vision-related quality of life:** improved in 21 of 23 participants (Michaelides *et al.*, 2023).
- **Visual acuity:** variable, modest improvements (Michaelides *et al.*, 2023).

Neurological Evidence of Cortical Cone Pathway Reactivation

A landmark companion study (Farahbakhsh *et al.*, 2022) used functional MRI population receptive field mapping combined with a 'silent substitution' psychophysical paradigm (stimuli isolating cone signals from rod responses) to probe whether gene therapy activated dormant cone pathways in visual cortex. In 4 paediatric patients (ages 10–15) drawn from both the CNGB3 and CNGA3 MeiraGTx trials, tested pre- and 6–14 months post-treatment, 2 of 4 children showed new cone-mediated retinotopic maps in visual cortex, expanded cortical coverage, and significantly improved cone contrast sensitivity thresholds — despite no improvement in best-corrected visual acuity. This was the first direct demonstration that retinal gene therapy can activate dormant cone photoreceptor pathways in the human brain, and that the visual cortex retains cone-mediated plasticity after up to 15 years without cone input (Farahbakhsh *et al.*, 2022).

What Does 'Completed' Mean? Did It Succeed?

The Phase 1/2 trial completed enrolment and follow-up and was published (Michaelides *et al.*, 2023). It met its primary safety endpoint and showed encouraging — though heterogeneous — efficacy signals, particularly for photo-aversion and quality of life. It did NOT progress to a Phase 3 trial. In December 2023, Johnson & Johnson Innovative Medicine fully acquired MeiraGTx's remaining interests in the achromatopsia programme. J&J subsequently decided NOT to advance AAV8-hCARp.hCNGB3 to further development — a strategic commercial decision, not a safety or efficacy failure (Rohowetz *et al.*, 2025). The companion long-term follow-up study (NCT03278873) was also terminated at this time.

Trial 2: AAV-CNGA3 — MeiraGTx (NCT03758404)

Status: Completed | **Phase:** 1/2 | **Sponsor:** MeiraGTx UK II Ltd (acquired by J&J 2023) | **Started:** August 2019

Background

Companion to NCT03001310, targeting the CNGA3 gene — the second most common cause of achromatopsia (Kohl *et al.*, 2005). In January 2021, the FDA granted Fast Track Designation for AAV-CNGA3, recognising the unmet medical need and preliminary biological evidence (MeiraGTx press release, 2021). 11 patients aged ≥3 enrolled.

Vector Design

Vector: **AAV2/8-hG1.7p.coCNGA3** — AAV8 capsid, modified cone-specific hG1.7 promoter, codon-optimised CNGA3 cDNA. Codon optimisation improves translational efficiency in human cells (Baxter and Borchert, 2024).

Results and Programme Status

The same Farahbakhsh *et al.* (2022) neuroimaging study included patients from both this CNGA3 trial and the CNGB3 trial. The cortical reactivation findings — new cone-mediated retinotopic maps in 2 of 4 paediatric patients — apply across both gene targets. Like the CNGB3 programme, this trial was fully acquired by J&J in December 2023 and development was discontinued on commercial grounds (Rohowetz *et al.*, 2025).

Trial 3: AGTC-401 (rAAV2tYF-PR1.7-hCNGB3) — Beacon Therapeutics (NCT02599922)

Status: Active, Not Recruiting | **Phase:** 1/2 | **Sponsor:** Beacon Therapeutics (acquired from AGTC, 2023) | **Started:** April 2016

Vector Design — Engineered Capsid

Vector: **rAAV2tYF-PR1.7-hCNGB3**. The rAAV2tYF capsid is AAV2 with surface tyrosine residues mutated to phenylalanine (Y→F), blocking phosphorylation-mediated ubiquitination and proteasomal degradation of the capsid, substantially increasing nuclear accumulation and transduction efficiency compared to wild-type AAV2 (AGTC press release, 2022). The PR1.7 promoter (~1.7 kb red cone opsin promoter fragment) drives cone-specific expression (AGTC press release, 2022).

Trial Design

21 adults and 10 paediatric patients enrolled across six doses; subretinal injection to one eye; up to 24 months follow-up (AGTC press release, 2022; ClinicalTrials.gov, 2026).

Results (February 2022 Interim Data)

AGTC reported robust improvements in visual sensitivity (AGTC press release, 2022):

- At the 1.1×10^{12} vg/mL dose: 2 of 4 paediatric patients and 2 of 3 adults classified as responders based on visual sensitivity and quality-of-life improvements (AGTC press release, 2022).
- Improvements in retinal sensitivity (static perimetry) and light discomfort at the second-highest dose (AGTC press release, 2022).
- **Safety:** 3 SUSARs (severe inflammation at highest doses), all resolving with corticosteroids; no permanent vision loss (AGTC press release, 2022).

AGTC announced plans in 2022 to continue AGTC-401 development. However, following Beacon Therapeutics' acquisition of AGTC in 2023, Beacon decided not to continue developing either achromatopsia programme commercially, prioritising its RPGR/XLRP programme instead (Rohowetz *et al.*, 2025; Foundation Fighting Blindness, 2023). The trial remains active for long-term safety follow-up of enrolled patients (ClinicalTrials.gov, 2026).

Trial 4: AGTC-402 (rAAV2tYF-PR1.7-hCNGA3) — Beacon Therapeutics (NCT02935517)

Status: Active, Not Recruiting | **Phase:** 1/2 | **Sponsor:** Beacon Therapeutics | **Started:** August 2017

Vector Design

Same rAAV2tYF capsid and PR1.7 promoter as AGTC-401, but carrying human CNGA3 cDNA. 16 adults and 8 children enrolled at five dose levels (AGTC press release, 2022; ClinicalTrials.gov, 2026).

Results — A Negative Trial

> Unlike AGTC-401, **AGTC-402 demonstrated no consistent evidence of biologic activity** across the dose levels tested (AGTC press release, 2022). No consistent improvements in visual function were observed. AGTC therefore discontinued development of AGTC-402 (AGTC press release, 2022). The reasons for this negative result in CNGA3 — while CNGA3 therapy showed functional signals in the MeiraGTx and STZ trials — are not fully resolved. Possible explanations include differences in vector design (capsid, promoter), codon optimisation status, or patient cohort characteristics. The discordance highlights the importance of vector design parameters, not just gene target, in determining efficacy (Baxter and Borchert, 2024).

Trial 5: rAAV8.hARR3p.hCNGA3 Bilateral — STZ eyetrial (NCT02610582)

Status: Active, Not Recruiting | **Phase:** 1/2 | **Sponsor:** STZ eyetrial (Univ. Hospital Tübingen / LMU Munich) | **Started:** November 2015

Background and Unique Features

This German academic trial, led by Drs Wissinger, Kohl, Michalakis and colleagues, was the first to explore **bilateral** subretinal delivery in achromatopsia — injecting both eyes rather than one. Most trials treat only the worse eye, but achromatopsia affects both eyes equally; bilateral treatment better addresses the disease burden (Reichel *et al.*, 2022).

Vector Design

Vector: **rAAV8.hARR3p.hCNGA3**. AAV8 capsid; human arrestin-3 (hArr3/ARR3) cone-specific promoter; human CNGA3 cDNA enhanced by a mutated WPRE for improved expression; AAV2 ITRs (Reichel *et al.*, 2022). Doses tested: 1×10^{10}, 5×10^{10}, and 1×10^{11} vector genomes across 9 adult patients in a dose-escalation design (Reichel *et al.*, 2022).

Three-Year Results (Reichel *et al.*, Br J Ophthalmol, 2022)

- **Safety (primary endpoint): Confirmed.** No adverse or serious adverse events related to the study drug occurred after year 1. Two patients had mild early inflammatory reactions resolving with topical corticosteroids (Reichel *et al.*, 2022).
- **Durability:** Functional improvements observed in the treated eye at year 1 were persistent at years 2 and 3 — confirming multi-year durability of AAV-mediated transgene expression in human photoreceptors (Reichel *et al.*, 2022).
- **Efficacy vs. untreated fellow eye:** While some secondary endpoints (e.g., cone ERG amplitude, some perimetry measures) reached statistical significance for the treated eye alone, comparisons between treated and untreated fellow eye did not reach statistical significance for most endpoints — a consequence of small sample size (n=9) and inherent outcome measure variability rather than absence of biological effect (Reichel *et al.*, 2022).

The trial is ongoing for long-term follow-up. As an academic programme, no commercial development decision has been announced.

Trial 6: NT-501 CNTF Implant — National Eye Institute (NCT01648452)

Status: Completed | **Phase:** 1/2 | **Sponsor:** National Eye Institute (NEI) | **Started:** July 2012

Background — Neuroprotection Rather Than Gene Replacement

This NIH trial tested a fundamentally different strategy: instead of replacing the defective gene, it investigated whether delivering ciliary neurotrophic factor (CNTF) — a cytokine with neuroprotective properties — could rescue dysfunctional cone photoreceptors. The rationale came from animal studies showing intraocular CNTF improved cone ERG responses in CNGB3-mutant dogs (Sieving *et al.*, 2014). Delivery was via the NT-501 encapsulated cell technology (ECT) implant: a small (~6 mm) silicone capsule containing ARPE-19 cells engineered to secrete CNTF at ~20 ng/day, placed surgically in the vitreous cavity (Sieving *et al.*, 2014). This is a pharmacological protein-delivery approach, not a gene therapy.

Trial Design and Results

5 participants with confirmed CNGB3 mutations and BCVA ≤20/100 received the implant in one eye; the fellow eye served as an untreated control; 1-year follow-up (Sieving *et al.*, 2014).

> CNTF did NOT improve cone function in any participant (Sieving *et al.*, 2014). Pupillary constriction data confirmed intraocular CNTF was being released. Unexpectedly, scotopic ERG responses were **reduced** in treated eyes, indicating CNTF suppressed rod photoreceptor function at the doses delivered. The authors concluded this represented a **species difference**: human CNGB3 cones do not respond to CNTF as canine cones do (Sieving *et al.*, 2014). This was a translational failure and reinforced the field's pivot toward gene replacement as the primary strategy. It also illustrates that successful animal models do not guarantee human efficacy.

Trial 7: Glycerol Phenylbutyrate (PBA) for ATF6-Mediated Achromatopsia — Columbia University (NCT04041232)

Status: Suspended | **Phase:** Early Phase 1 | **Sponsor:** Columbia University | **Started:** April 2025

ATF6 Mechanism and PBA Rationale

ATF6 mutations (~2% of achromatopsia) impair the ER's UPR capacity, causing unresolved ER stress and cone degeneration (Ansar *et al.*, 2015). Glycerol phenylbutyrate (PBA; trade name Ravicti) is an FDA-approved nitrogen scavenger for urea cycle disorders. Phenylbutyrate acts as a **chemical chaperone**: it non-specifically promotes protein folding stability in the ER, reducing the burden of misfolded proteins and thereby alleviating ER stress — partially substituting for the absent ATF6-mediated UPR (ClinicalTrials.gov, 2026; Baxter and Borchert, 2024). Preclinical data from the Columbia group demonstrated that PBA improved retinal function in ATF6-/- mice (ClinicalTrials.gov, 2026).

Trial Design and Current Status

Patients receive oral PBA 3× daily at 4.5–11.2 mL/m²/day (≤17.5 mL/day), monitored over ≥3 clinic visits with ERG, visual acuity, OCT and fundoscopy (ClinicalTrials.gov, 2026). The trial is currently listed as **Suspended**; reasons not publicly disclosed. This represents a non-GT drug repurposing approach for a rare subgroup — relevant as a contrast to the gene replacement paradigm dominating the field.

5. Observational / Natural History / Registry Studies

Natural history studies provide the baseline phenotypic characterisation, patient identification, and longitudinal disease course data required to design and interpret interventional trials (Baxter and Borchert, 2024).

NCT04124185 — Natural History Study, MeiraGTx (Completed from 2011)

Initiated in advance of MeiraGTx's Phase 1/2 trials, using OCT, adaptive optics scanning light ophthalmoscopy (AOSLO), microperimetry and ERG to characterise cone photoreceptor structure and function in achromatopsia patients (ClinicalTrials.gov, 2026). OCT findings in achromatopsia include disruption of the EZ at the fovea; AOSLO can resolve individual cone inner segments, providing direct evidence of surviving target cells for gene therapy. Degree of EZ integrity correlates with functional severity and likely predicts gene therapy responsiveness (Genead *et al.*, 2011; Rohowetz *et al.*, 2025). These data informed patient eligibility criteria and outcome measure selection for the MeiraGTx interventional trials.

NCT01846052 — Clinical and Genetic Characterisation, Beacon/AGTC (Completed from 2013)

AGTC's parallel natural history study: CNGB3 achromatopsia patients followed every 6 months for up to 1.5 years, establishing a phenotypic database and identifying candidates for the AGTC-401 interventional trial (ClinicalTrials.gov, 2026).

NCT07085533 — IRD Natural History, Zhongmou Therapeutics (Recruiting from July 2025)

~200 IRD patients and controls examining chromatic vision deficits versus structural biomarkers (EZ integrity on SD-OCT), and validating the Moji Low-Vision Color Discrimination Test for patients with severely reduced acuity (ClinicalTrials.gov, 2026).

NCT02435940 — My Retina Tracker Registry, Foundation Fighting Blindness (Recruiting since 2014)

Patient-initiated global registry (>40,000 enrolled with inherited retinal diseases) capturing genotypic/phenotypic data, genetic test results and clinical measurements. Used to accelerate trial recruitment and establish population-level genotype-phenotype correlations (ClinicalTrials.gov, 2026; Foundation Fighting Blindness, 2023).

NCT06491615 — eyeGENE Stage 3, NIH (Recruiting since July 2024)

Third stage of the NIH eyeGENE biobank: targeted accrual of DNA samples and clinical data from patients with rare inherited eye diseases including achromatopsia, supporting genetic research and future trial recruitment (ClinicalTrials.gov, 2026).

6. Why These Approaches? Subretinal Delivery, AAV8 and Cone Promoters

Why Subretinal Not Intravitreal

The inner limiting membrane (ILM) substantially blocks AAV penetration from the vitreous to the photoreceptor layer; standard AAV8 intravitreal delivery primarily transduces inner retinal cells (ganglion cells, Müller glia) rather than cones (Baxter and Borchert, 2024). Subretinal placement positions the vector in direct contact with cone outer segments and the immunologically privileged subretinal space, dramatically improving transduction efficiency and reducing the vector dose required (Baxter and Borchert, 2024; Reichel *et al.*, 2022).

Why AAV8, and What Makes the Capsid Choice Matter

AAV8 achieves superior cone photoreceptor transduction compared to AAV1, 2, 4 or 6 after subretinal delivery, uncoats faster in the nucleus, and has lower pre-existing immunity in humans than AAV2 (Baxter and Borchert, 2024). AGTC's rAAV2tYF approach achieved AAV2-like receptor binding (familiar tropism) while blocking capsid degradation, improving nuclear delivery of AAV2 vector to levels approximating or exceeding AAV8 (AGTC press release, 2022).

Why Cone-Specific Promoters Are Essential

CNG channels are functional ion channels — ectopic expression in non-cone cells could cause abnormal electrophysiology or toxicity. Cone-specific promoters (hCAR, ARR3, PR1.7) restrict expression to the therapeutic target, improve the safety profile, and concentrate the limited protein production capacity of transduced cells on producing therapeutic levels in the right location (Baxter and Borchert, 2024; Michaelides *et al.*, 2023).

The Therapeutic Window

The degree of preserved cone inner segment structure on OCT (EZ integrity) is likely a stronger predictor of gene therapy response than age alone (Genead *et al.*, 2011; Rohowetz *et al.*, 2025). The neuroimaging findings of Farahbakhsh *et al.* (2022) suggest that treating in childhood — while cortical visual pathways retain plasticity — may produce superior functional gains than treating in adulthood, even where cone structure is preserved. This has prompted paediatric cohort expansions in both the MeiraGTx and AGTC trials.

7. Trial Summary Table

NCT	Vector / Drug	Gene	Ph	Status	Outcome Summary (key citations)
NCT03001310	AAV8-hCARp.hCNGB3 MeiraGTx	CNGB3	1/2	Completed	Safe (3/23 inflammation). Photo-aversion 11/20; QoL 21/23. fMRI: new cortical cone maps in 2/4 paeds. J&J; discontinued commercially (Michaelides et al., 2023; Farahbakhsh et al., 2022; Rohowetz et al., 2025).
NCT03758404	AAV8-hG1.7p.coCNGA3 MeiraGTx	CNGA3	1/2	Completed	FDA Fast Track 2021. Same fMRI study showed cortical reactivation (Farahbakhsh et al., 2022). J&J; discontinued commercially (Rohowetz et al., 2025).
NCT02599922	rAAV2tYF-PR1.7-hCNGB3 AGTC-401	CNGB3	1/2	Active, NR	Robust visual sensitivity gains in 2/4 paeds and 2/3 adults at top dose. 3 SUSARs, resolved. Beacon discontinued programme (AGTC press release, 2022; Rohowetz et al., 2025).
NCT02935517	rAAV2tYF-PR1.7-hCNGA3 AGTC-402	CNGA3	1/2	Active, NR	No consistent biologic activity. Development discontinued by Beacon/AGTC (AGTC press release, 2022).
NCT02610582	rAAV8.hARR3p.hCNGA3 bilateral STZ	CNGA3	1/2	Active, NR	Safety confirmed (primary). Durable 3-yr functional benefits in treated eye; vs. fellow eye mostly non-significant (small n=9) (Reichel et al., 2022).
NCT01648452	NT-501 CNTF implant NEI	CNGB3 (neuroprot.)	1/2	Completed	NEGATIVE. No cone improvement; rod ERG suppressed. Species difference from dog. Reinforced gene replacement approach (Sieving et al., 2014).
NCT04041232	Glycerol PBA oral Columbia	ATF6	E.Ph 1	Suspended	Preclinical: PBA improved ATF6-/- mouse retinas. Trial suspended, reason undisclosed. Non-GT drug repurposing for rare ATF6 subtype (ClinicalTrials.gov, 2026).

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